Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A satellite is revolving in a circular orbit at a height h above the surface of the earth of radius R. The speed of the satellite in its orbit is one-fourth the escape velocity from the surface of the earth. The relation between h and R is,

A. $h = 3R$

B. $h = 5R$

C. $h = 7R$

D. $h = 4R$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A child is standing with folded hands at the centre of a platform rotating about its central axis. The kinetic energy of the system is K. The child now stretches his arms so that the moment of inertia of the system doubled. The kinetic energy of the system now is

A. $2K$

B. $\frac{K}{2}$

C. $\frac{K}{4}$

D. $4K$

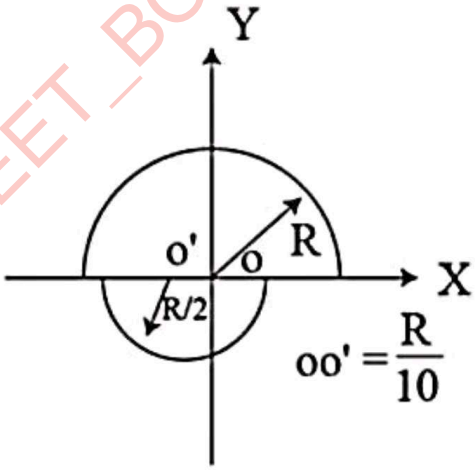
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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two solid hemispheres of radii R and $\frac{R}{2}$ with centers O and O' respectively as shown in figure. The density of bigger hemisphere is ρ and that of smaller hemisphere is 2ρ. Taking center of bigger hemisphere is at origin and the distance between centers of two hemispheres OO' is $\frac{R}{10}$ find coordinates of center of mass of the system.



A. $\left(-\frac{R}{50}, \frac{21R}{80}\right)$

B. $\left(-\frac{R}{30}, \frac{21R}{80}\right)$

C. $\left(-\frac{R}{50}, \frac{7R}{16}\right)$

D. $\left(-\frac{R}{30}, \frac{7R}{16}\right)$

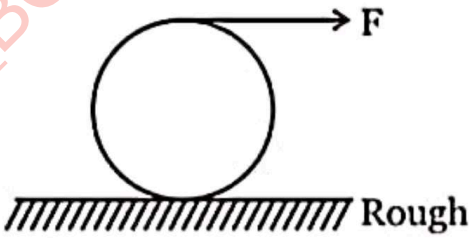
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Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solid sphere of radius R and mass M is rolled by a force F acting at the top of the sphere as shown in the figure. The sphere starts from rest and rolls without slipping, then



A. work done by force F when the center moves a distance S is $2FS$

B. work done by force F when the center moves a distance S is $3FS$

C. work done by force F when the center moves a distance S is FS

D. work done by force F when the center moves a distance S is $4FS$

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}\,g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 km/s, the escape speed on the surface of the planet in km/s will be

A. $5\ \text{km s}^{-1}$

B. $4\ \text{km s}^{-1}$

C. $3\ \text{km s}^{-1}$

D. $1\ \text{km s}^{-1}$

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A rigid body is in pure rotation, that is undergoing fixed axis rotation. Then which of the following statements are true -

A. You can find two points in the body in a plane perpendicular to the axis of rotation having same velocity

B. You can find two points in the body in a plane perpendicular to the axis of rotation having same acceleration

C. Speed of all the particles lying on the curved surface of a cylinder whose axis coincides with the axis of rotation is same

D. none of these

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : Kepler's second law can be understood by conservation of angular momentum principle.
Reason : Kepler's second law is related with areal velocity which can further be proved to be based on conservation of angular momentum as $(dA/dt) = (r^2\omega)/2$.

A. If both assertion and reason are true and reason is the correct explanation of assertion.

B. If both assertion and reason are true but reason is not the correct explanation of assertion.

C. If assertion is true but reason is false.

D. If both assertion and reason are false.

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A thin rod of mass m hinged at its upper end is to be maintained in equilibrium, making an angle θ with the vertical. The minimum force that needs to be applied to its bottom end to make this possible is :



A. $\frac{mg}{2\sin\theta}$

B. $\frac{mg}{\cos\theta}$

C. $\frac{mg\sin\theta}{2}$

D. $mg\sin\theta$

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let ω is the angular velocity of the earth's rotation about its own axis and acceleration due to gravity on the earth's surface at the poles is g. An object weighed by a spring balance gives the same reading at the equator, as well as at a height h above the poles $(h \ll R)$. What is the value of h ?

A. $\frac{gR}{\omega}$

B. $\frac{\omega^2 R^2}{g}$

C. $\frac{\omega^2 R^2}{2g}$

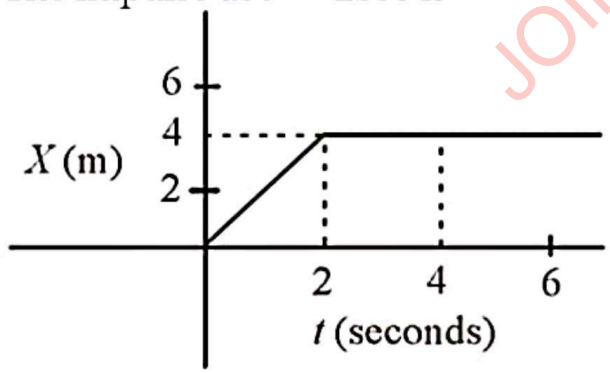
D. $\frac{2\omega^2 R^2}{g}$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the figure given, the position-time graph of a particle of mass 0.1 kg is shown.
The impulse at $t = 2$sec is



A. 0.2 kg msec^{-1}

B. -0.2 kg msec^{-1}

C. 0.1 kg msec^{-1}

D. -0.4 kg msec^{-1}

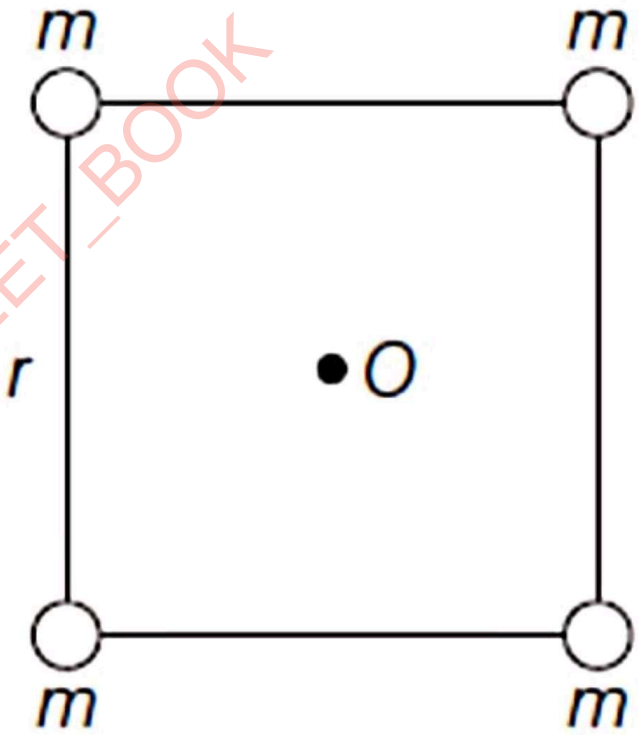
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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

There is a system of four mass particles placed (fixed) at the edge of a square of side r. The workdone to bring a mass m from infinity to the centre of square is



A. $2\sqrt{2}Gm^2/r$

B. $2\mathrm{Gm}^2/\mathrm{r}$

C. $\frac{4\sqrt{2}Gm^2}{r}$

D. $\frac{4Gm^2}{r}$

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A block of mass $m_1 = 100g$ rests on a horizontal table. The coefficient of friction between the block and the table is 0.05. When hit by another mass $m_2 = 100g$ moving with speed v that gets embedded in it, the combined mass moves and comes to stop after moving a distance of 4 m on the table. If a freely falling object were to acquire speed $v/2$ after being dropped from height H then neglecting energy losses and taking $g = 10\ \text{ms}^{-2}$, the value of H is close to:

A. 0.3 m

B. 0.2 m

C. 0.4 m

D. 0.25 m

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: The artificial satellite does not have any fuel but even then it remains orbiting around the earth.

Reason: The necessary centripetal force required to move the satellite in an orbit around the earth is provided by the gravitational force of attraction between the satellite and the earth.

A. Assertion is true, reason is true; reason is a correct explanation for assertion.

B. Assertion is true, reason is true; reason is not a correct explanation for assertion.

C. Assertion is true, reason is false.

D. Assertion is false, but reason is true.

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A wheel of moment of inertia 2.5 Kg-m^2 has an initial angular velocity of 40 rads $^{-1}$. A constant torque of 10 Nm acts on the wheel. The time during which the wheel is accelerated to 60 rads^{-1} is

A. 6 s

B. 5 s

C. 4.5 s

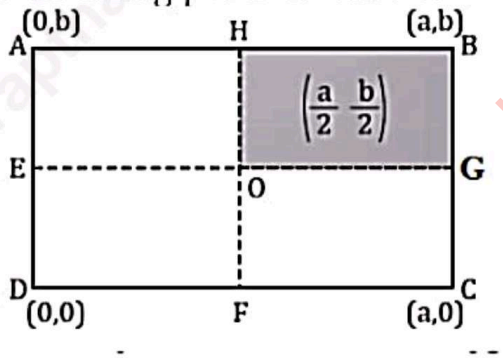
D. 4 s

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be:



A. $\frac{5a}{12}, \frac{5b}{12}$

B. $\frac{5a}{3}, \frac{5b}{3}$

C. $\frac{2a}{3}, \frac{2b}{3}$

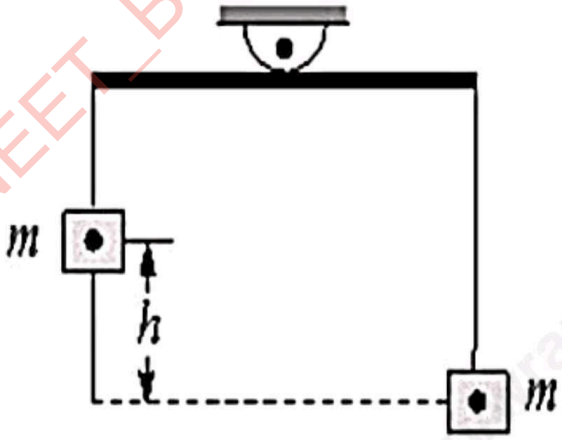
D. $\frac{3a}{4}, \frac{3b}{4}$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two identical blocks of mass m are suspended from a beam balance whose scale pans differ in vertical height by $h(\ll R)$, If R and ρ are the radius and density of the earth, then the error in weighing is



A. $\frac{2}{3}\pi\rho R^3 Gm$

B. $\frac{8}{3}\pi\rho Gmh$

C. $\frac{8}{3}\pi\rho R^3 Gm$

D. $\frac{4}{3}\pi\rho Gm^2 h$

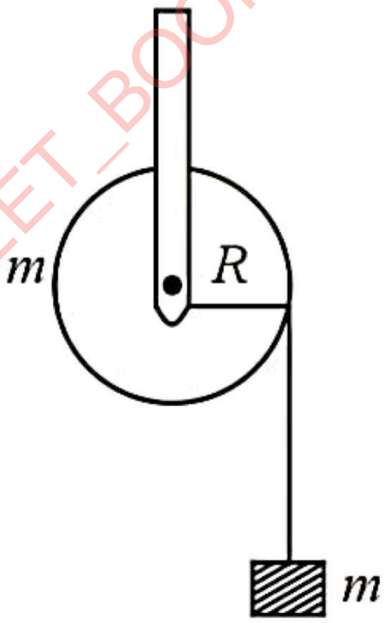
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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A mass 'm' supported by a massless string wound around a uniform solid cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release?



A. $\frac{2\,g}{3}$

B. $\frac{g}{2}$

C. $\frac{5\,g}{6}$

D. g

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : In an elastic collision of two billiard balls, the total kinetic energy is conserved during the short time of oscillation of the balls (i.e. when they are in contact).
Reason : Energy spent against friction does not follow the law of conservation of energy.

A. If both the assertion and reason are true and reason is a correct explanation of the assertion.

B. If both assertion and reason are true but assertion is not a correct explanation of the assertion.

C. If the assertion is true but the reason is false.

D. If both assertion and reason are false.

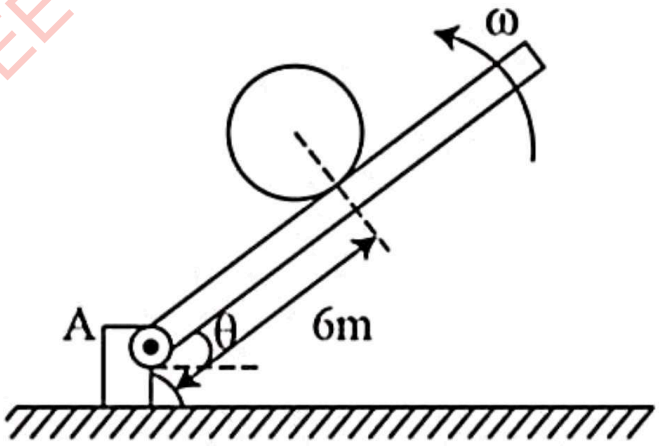
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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A cylinder weighing 450 N with a radius of 30 cm is held fixed on an incline that is rotating at $0.5 \text{rad} s^{-1}$. The cylinder is released when the incline is at position θ equal to 30°. If the cylinder is 6 m from the bottom A at the instant of release, what is the initial acceleration of the centre of the cylinder relative to the incline, if there is no slipping? $\left(\text{g} = 10 \text{ m s}^{-2}\right)$



A. 2.33 m s^{-2}

B. 4.66 m s^{-2}

C. 1.33 m s^{-2}

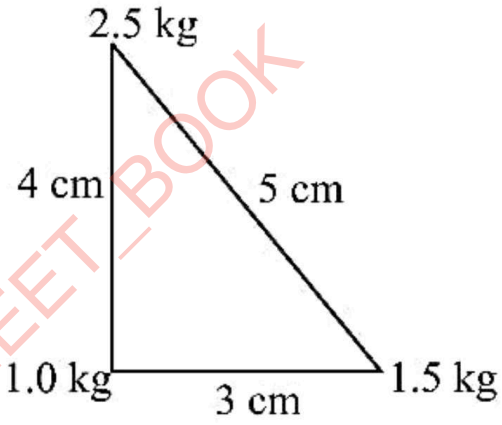
D. 3.33 m s^{-2}

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three point particles of masses 1.0 kg, 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm, 3.0 cm and 5.0 cm as shown in the figure. The centre of mass of the system is at a point:



A. 0.6 cm right and 2.0 cm above 1 kg mass.

B. 1.5 cm right and 1.2 cm above 1 kg mass.

C. 2.0 cm right and 0.9 cm above 1 kg mass.

D. 0.9 cm right and 2.0 cm above 1 kg mass.

Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

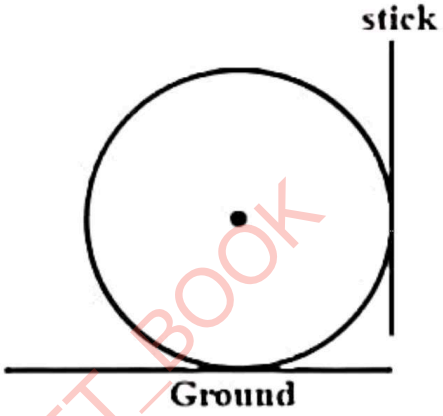
A rocket has to be launched from the Earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon is found to be $\frac{E}{N}$. What is the value of N ? [Note: Assume $\rho_{\mathrm{m}} = \rho_{\mathrm{e}}$; Volume of Earth $= 64 \times$ Volume of the moon]

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

A boy is pushing a ring of mass $2\ \mathrm{kg}$ and radius $0.5\ \mathrm{m}$ with a stick as shown in the figure. The stick applies a force of $2\ \mathrm{N}$ on the ring and rolls it without slipping with an acceleration of $0.3\ \mathrm{m\ s^{-2}}$. The coefficient of friction between the ground and the ring is large enough so that rolling always occurs. The coefficient of friction between the stick and the ring is $P/10$. The value of P is

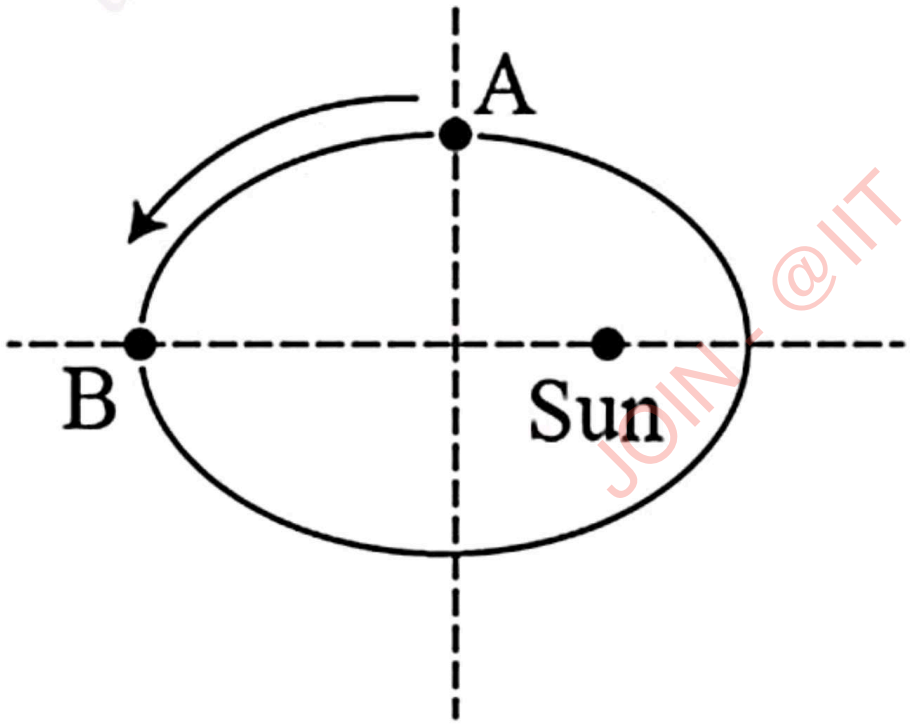


Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A planet is revolving around the sun in an elliptical orbit having eccentricity $e\left(=\frac{\pi}{4}\right)$. If the time period of revolution is 16 months then find the time taken by the planet in going from A to B in months.



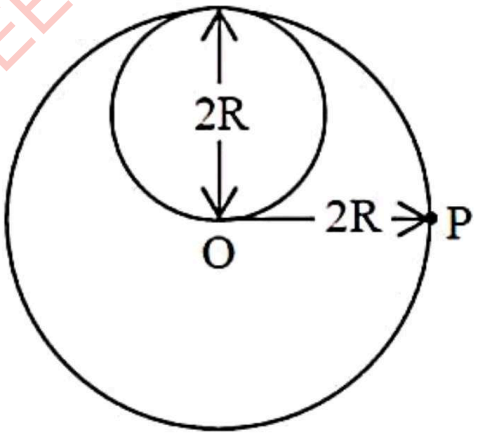
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Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A lamina is made by removing a small disc of diameter $2R$ from a bigger disc of uniform mass density and radius $2R$, as shown in the figure. The moment of inertia of this lamina about axes perpendicular to the plane of the lamina and passing through the points O and P is I_O and I_P respectively. If the ratio $\frac{I_P}{I_O} = \frac{m}{n}$, where m and n are the smallest integers, then what is the value of $m + n$?



Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A gun of mass 10 kg fires 4 bullets per second. The mass of each bullet is 20 g and the velocity of the bullet when it leaves the gun is 300 m s^{-1}. The force required to hold the gun when firing is

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solubility product values of barium sulphate and barium carbonate are 1.0×10^{-10} and 5.0×10^{-9} respectively. When the two salts are equilibrated in pure water the ratio of $\frac{[SO_4^{2-}]}{[CO_3^{2-}]}$ is

A. 0.02

B. 0.10

C. 0.05

D. 0.12

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Question No. 2

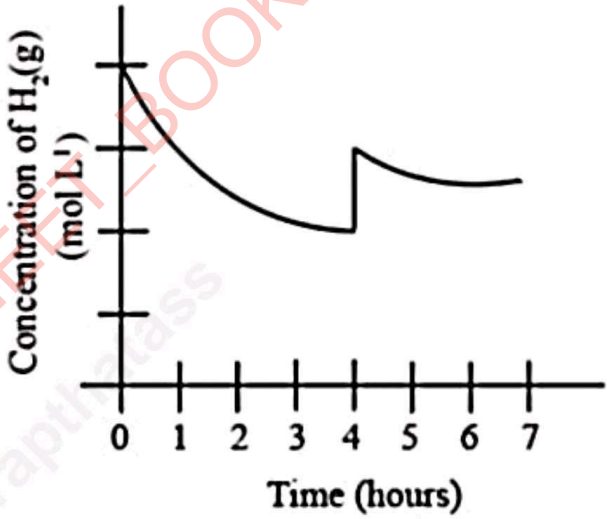
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The reaction between phosphorus (P_4) and hydrogen (H_2) can result in the formation of phosphine (PH_3) as shown:

$$P_4(s) + 6H_2(g) \rightleftharpoons 4PH_3(g)$$

The graph shows the change in concentration of hydrogen for this reaction in which the system was disturbed after four hours.



Which of the following could explain the change in hydrogen concentration at time = 4 hours?

A. The pressure on the reaction mixture was increased

B. A catalyst was added

C. The volume on the reaction mixture was decreased

D. More phosphorus was added

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The enthalpy of formation of $CO_{(g)}$, $CO_{2(g)}$, $N_2O_{(g)}$ and $N_2O_{4(g)}$ is $-110, -393, +811$ and $10\ kJ/mol$ respectively. For the reaction,
$N_2O_{4(g)} + 3CO_{(g)} \rightarrow N_2O_{(g)} + 3CO_{2(g)}; \Delta H_r(\ kJ/mol)$ is

A. -212

B. +212

C. +48

D. -48

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What will be the pH of a solution by mixing 40 mL of 0.1 M HCl with 10 mL of 0.45 M NaOH ?

A. 12

B. 10

C. 8

D. 6

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the following

Column I	Column II
(a) Enthalpy	(p) Intensive property
(b) Temperature	(q) Path function
(c) Free energy	(r) Function of U, P and V
(d) Work	(s) State function

A. $a \rightarrow r, s; b \rightarrow p, s; c \rightarrow r, s; d \rightarrow q$

B. $a \rightarrow p; b \rightarrow r; c \rightarrow q; d \rightarrow r$

C. $a \rightarrow r; b \rightarrow q; c \rightarrow r; d \rightarrow p$

D. $a \rightarrow q; b \rightarrow r; c \rightarrow p; d \rightarrow r$

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

4ml of HCl solution of $pH = 2$ is mixed with 6ml of NaOH solution of $pH = 12$. What would be the final pH of solution? $(\log 2 = 0.3)$

A. 10.3

B. 11.3

C. 11

D. 4.3

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a flask containing 0.5 atm pressure of $A_2(g)$, some solid AB is added which undergoes dissociation according to $2AB(s) \rightleftharpoons A_2(g) + B_2(g), K_p = 0.06\ \text{atm}^2$
The total pressure (atm) at equilibrium will be

A. 0.5 atm

B. 0.2 atm

C. 0.7 atm

D. 1.4 atm

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given:

$$\text{C(graphite)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})\ ;\ \Delta_\text{r}\text{H}^\circ = -393.5\ \text{kJ mol}^{-1}$$

$$\text{H}_2(\text{g}) + \tfrac{1}{2}\,\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O(l)};\ \Delta_\text{r}\text{H}^\circ = -285.8\ \text{kJ mol}^{-1}$$

$$\text{CO}_2(\text{g}) + 2\text{H}_2\text{O(l)} \rightarrow \text{CH}_4(\text{g}) + 2\text{O}_2(\text{g});\ \Delta_\text{r}\text{H}^\circ = +890.3\ \text{kJ mol}^{-1}$$

Based on the above thermochemical equations, the value of $\Delta_\text{r}\text{H}^\circ$ at 298 K for the reaction

$\text{C(graphite)} + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$ will be:

A. $+144.0\ \text{kJ mol}^{-1}$

B. $-74.8\ \text{kJ mol}^{-1}$

C. $-144.0\ \text{kJ mol}^{-1}$

D. $+74.8\ \text{kJ mol}^{-1}$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solution is 0.10 M $Ba(NO_3)_2$ and 0.10 M $Sr(NO_3)_2$. If solid, Na_2CrO_4 is added to the solution, what is $\left[Ba^{2+}\right]$ when $SrCrO_4$ begins to precipitate?
$\left[K_{sp}(BaCrO_4) = 6 \times 10^{-9}, K_{sp}(SrCrO_4) = 7 \times 10^{-6}\right]$

A. 8.5×10^{-5}M

B. 9.5×10^{-5}M

C. 8.5×10^{-9}M

D. 8.5×10^{-6}M

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Among the following, the reaction that is accompanied by a decrease in the entropy is

A. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

B. $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$

C. $PCl_5(s) \rightarrow PCl_3(l) + Cl_2(g)$

D. $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the decomposition reaction of N_2O_4 the equilibrium constant Kp is 0.120 atm at $298K$ and the total pressure of system is 2 atm. What will be the value of degree of dissociation of N_2O_4?

A. 0.120

B. 0.242

C. 1.21

D. 0.212

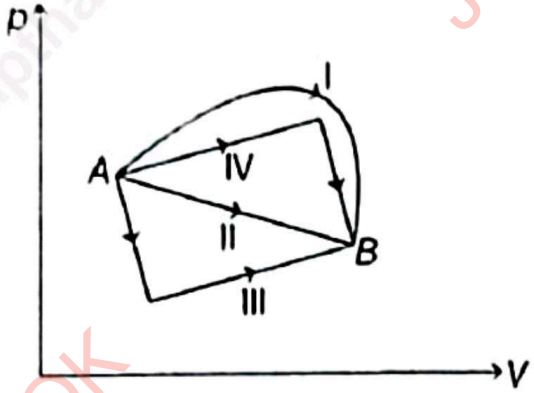
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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

From the following p-V diagram, an ideal gas undergoing a change of state from A to B. Four different processes I, II, III and IV as shown in the figure may lead to same change of state.



A. Work done is minimum in case II

B. Change in internal energy is same in all the four cases

C. Change in internal energy is same in IV and III cases, but not in I and II

D. None of these

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The ratio of volumes of CH_3COOH 0.1 (N) to CH COONa 0.1 (N) required to prepare a buffer solution of pH 5.74 is (Given, pK_a of CH_3COOH is 4.74)

A. $10:1$

B. $5:1$

C. $1:5$

D. $1:10$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three samples of 100 g of water (samples I, II and III), initially kept at 1 atm pressure and 298 K were given the following treatments.

Sample I was heated to 320 K and cooled to 298 K

Sample II was heated to 300 K, cooled to 273 K and heated to 298 K

Sample III was heated to 373 K and cooled to 298 K

At the end of these processes, the internal energy of

A. III is the highest

B. II is the highest

C. I and III are the same, II is lower than that of I and III

D. I, II and III are the same

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following is strongest Bronsted base?

A. ClO^-

B. ClO_2^-

C. ClO_3^-

D. ClO_4^-

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the following statements.
I. The entropy in isolated system with only $p - V$ work is always maximised at equilibrium.
II. It is possible for the entropy of close system to increase substantially in an irreversible process.
III. Without the help of an external agency, spontaneous process cannot be reversed.
IV. ΔS_{system} is zero for reversible process in an isolated system.
Which of the above statement(s) is/are correct?

A. I, II, III

B. I, IV

C. II and III

D. All

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : With change in concentration of the reactant, the value of equilibrium constant changes.

Reason : More the value of K_C lesser is the extent of reaction in the forward direction.

A. If both Assertion & Reason are true and the reason is the correct explanation of the assertion, then mark (1).

B. If both Assertion & Reason are true but the reason is not the correct explanation of the assertion, then mark (2).

C. If Assertion is true statement but Reason is false, then mark (3).

D. If both Assertion and Reason are false statements, then mark (4).

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which one of the following is not a buffer solution?

A. $0.8M\ H_2\ S + 0.8M\ KHS$

B. $2M\ C_6H_5NH_2 + 2M\ C_6H_5NH_3{}^+Br^-$

C. $3M\ H_2CO_3 + 3M\ KHCO$

D. $0.05M\ KClO_4 + 0.05M\ HClO_4$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The standard entropy of X_2Y_2 and XY_3 are 60, 40 and 50 JK^{-1} mol^{-1}, respectively. For the reaction $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -30$ kJ, to be at equilibrium the temperature will be

A. 1000 K

B. 1250 K

C. 500 K

D. 750 K

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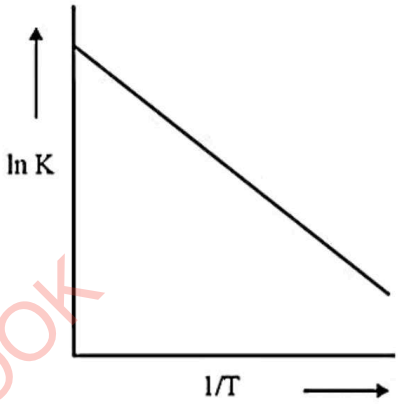
Question No. 20

Only One Option Correct Type

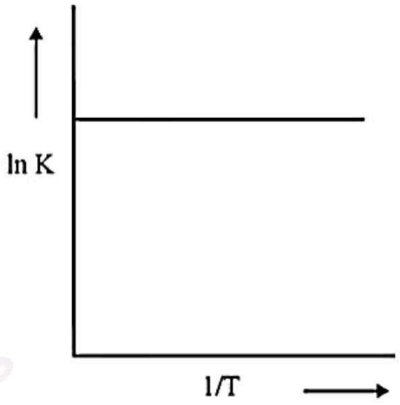
Each question has multiple options out of which ONLY ONE is correct.

Which In K vs $1/T$ plot is correct for an equilibrium that shifts toward reactants at higher temperatures?

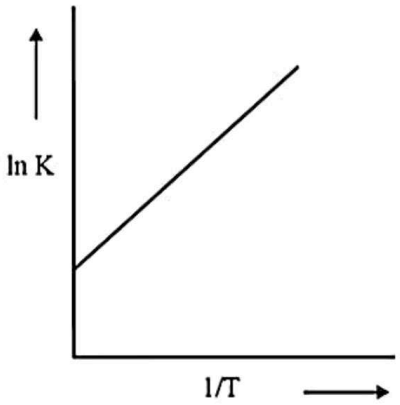
A.



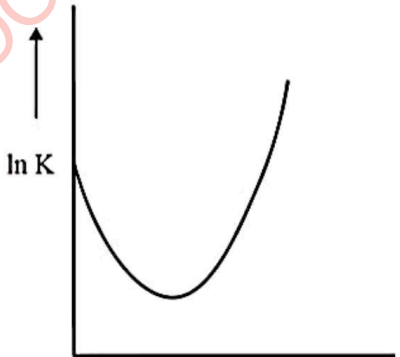
B.



C.



D.



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Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The dissociation constant of a substituted benzoic acid at $25^{\circ}\mathrm{C}$ is 1.0×10^{-4}. The pH of a $0.01\mathrm{M}$ solution of its sodium salt is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

$\mathrm{CaCO_{3(\,s)}} \rightleftharpoons \mathrm{CaO_{(s)}} + \mathrm{CO_{2(g)}}; \mathrm{K_p} = 1.64$ atm at 1000 K
50 g of $\mathrm{CaCO_3}$ in a 10 litre closed vessel is heated to 1000 K.
Percentage of $\mathrm{CaCO_3}$ that remains unreacted at equilibrium is :
(Given $\mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$)

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

If one monoatomic gas is expanded adibatically from 2 L to 10 L at 1 atm external pressure then the value of ΔU (in atm L) is (mark magnitude of internal energy change as answer)

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The standard enthalpies of formation of $\mathrm{Al_2O_3}$ and CaO are $-1675\ \mathrm{kJ\ mol^{-1}}$ and $-635\ \mathrm{kJ\ mol^{-1}}$ respectively.

For the reaction

$3\mathrm{CaO} + 2\mathrm{Al} \rightarrow 3\mathrm{Ca} + \mathrm{Al_2O_3}$ the standard reaction enthalpy $\Delta_{\mathrm{r}}\mathrm{H}^0 =$ _____ kJ.

(Round off to the Nearest Integer).

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Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2?

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the letters of the word 'INCURSION'. Number of arrangements of all letters in which vowels are in alphabetical order, is

A. $\frac{9!}{2!^2\ 4!}$

B. $\frac{9!}{2!\ 4!}$

C. ${}^9C_4$

D. ${}^9P_5$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of positive integral solutions of the inequality $x + y + z \leq 20$ is

A. 1008

B. 1028

C. 1108

D. 1140

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let S_1 is the complete solution set of the inequality $\cos^{-1}(x) > \cos^{-1}\left(x^2\right)$ and S_2 is the complete solution set of the inequality $\left(\cot^{-1}x\right)^2 - 5\cot^{-1}x + 6 > 0$, then $S_1 \cap S_2$ is

A. $(-1, 0)$

B. $(\cot 3, 0)$

C. $(\cot 2, 0)$

D. $(-1, \cot 2)$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If a 3-digit number is randomly chosen, what is the probability that either the number itself or some permutation of the number (which is a 3 -digit number) is divisible by 4 and 5 ?

A. $\frac{1}{45}$

B. $\frac{29}{180}$

C. $\frac{11}{60}$

D. $\frac{1}{4}$

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The sides AB, BC and CA of a ΔABC have a, b and c interior points on them respectively, then the number of triangles that can be constructed using these interior points as vertices, is

A. $\Sigma ab(a+b-2)$

B. $\Sigma ab(a+b)$

C. $\sum a^2$

D. None of These

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $\sum_{r=1}^{\infty} \cot^{-1}\left(3r^2 - r - \frac{1}{3}\right)$ is equal to

A. $\cot^{-1}1$

B. $\cot^{-1}\sqrt{3}$

C. $\cot^{-1}0$

D. $\cot^{-1}(-1)$

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three persons A, B, C throw a die in succession. The one getting 'six' wins. If A starts then the probability of B winning is 30/p, then what is the value of p.

A. 102

B. 46

C. 73

D. 91

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A box contains balls of two different colours blue and green. There are 3 blue balls. If two balls are to be drawn, together at random from the box, the probability that there is one ball of each colour is $\frac{1}{2}$. Possible number of green balls in box can be _____.

A. 6

B. 3

C. 8

D. 7

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of ways in which three numbers in arithmetic progression can be selected from $\{1, 2, 3, \ldots\ldots, 50\}$ is

A. 276

B. 600

C. 840

D. 640

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If letters of the word $REGULATIONS$ are randomly arranged, then the probability that there are exactly four letters between R and S is

A. $\frac{6}{55}$

B. $\frac{3}{55}$

C. $\frac{5}{11}$

D. $\frac{6}{11}$

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The sum of all 4-digit numbers that can be formed by using the digits $2, 4, 6, 8$ (repetition of digits is not allowed) is

A. 133320

B. 123330

C. 113230

D. 323430

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let A and B be two events such that $P(A \cap B) = \frac{1}{6}$, $P(A \cup B) = \frac{31}{45}$ and $P(\bar{B}) = \frac{7}{10}$ then

A. A and B are independent

B. A and B are mutually exclusive

C. $P\left(\frac{A}{B}\right) < \frac{1}{6}$

D. $P\left(\frac{B}{A}\right) < \frac{1}{6}$

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If N is the number of positive integral solutions of the equation $x_1x_2x_3x_4 = 770$, then the value of N is

A. 250

B. 252

C. 254

D. 256

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A continuous even periodic function f with period 8 is such that $f(0) = 0, f(1) = -2, \quad f(2) = 1, \quad f(3) = 2, \quad f(4) = 3$ then the value of $\tan^{-1}\left(\tan\left(f(-5) + f(20) + \cos^{-1}(f(-10) + f(17))\right)\right)$ is equal to:

A. $2\pi - 5$

B. $5 - 2\pi$

C. $3 + \pi$

D. $3 - \pi$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The letters of the word '$ASSASSIN$' are written down at random in a row. The probability that no two S occur together is

A. $\frac{1}{35}$

B. $\frac{1}{14}$

C. $\frac{1}{15}$

D. None of these

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of 6 -digit numbers of the form ababab (in base 10) each of which is a product of exactly 6 distinct primes is

A. 8

B. 10

C. 13

D. 15

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Box 1 contains 30 cards numbered 1 to 30 and Box 2 contains 20 cards numbered 31 to 50. A box is selected at random and a card is drawn from it. The number on the card is found to be a non-prime number. The probability that the card was drawn from Box 1 is

A. $\frac{2}{3}$

B. $\frac{8}{17}$

C. $\frac{4}{17}$

D. $\frac{2}{5}$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of even proper divisors of 1008 is

A. 23

B. 22

C. 25

D. None of these

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of the expression
$\sec^2\left(\tan^{-1}2\right) + \operatorname{cosec}^2\left(\cot^{-1}3\right) + \operatorname{cosec}\left(2\cot^{-1}2 + \cos^{-1}\frac{3}{5}\right)$
is equal to:

A. $15 + \frac{24}{25}$

B. $24 + \frac{15}{25}$

C. $25 + \frac{15}{24}$

D. $15 + \frac{25}{24}$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

2 dice are thrown. Suppose a random variable X is assigned a value $2k$, if the sum on the dice is equal to k, then the expected value of X is

A. 10

B. 12

C. 14

D. $\frac{50}{9}$

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Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Given two independent events, if the probability that cxactly onc of them occurs is $\frac{26}{49}$ and the probability that none of them occurs is $\frac{15}{49}$, then the probability of more probable of the two events is: x, then the value of 49x is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The number of ways in which 8 non-identical apples can be distributed among 3 boys such that every boy should get atleast 1 apple and atmost 4 apples is $k \cdot P_3^7$, where k has the value equal to

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

If a_1, a_2, b_1 and b_2 take values in the set $\{1, -1, 0\}$, then the probability that the equation $a_1 a_2 = b_1 b_2$ is satisfied is $\frac{p}{q}$, (p & q are co-prime) then, $q - 2p$ is

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Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

If the maximum value of x which satisfies the inequality $\sin^{-1}(\sin x) \geq \cos^{-1}(\sin x)$ for $x \in \left(\frac{\pi}{2}, 2\pi\right)$ is λ, then $\frac{2\lambda}{3}$ is equal to $\frac{\pi}{a}$ then a is equal to

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Number of ways in which 10 identical balls can be distributed in 5 boxes labeled $A,\ B,\ C,\ D$ & E such that no two adjacent box are empty together is equal to k. Then sum of digits of $\left[\frac{k}{11}\right]$ is equal to (where [.] denotes greatest integer function)

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